

DSA-110 Continuum Imaging

Where the pipeline stands · July 2026

Status for science collaborators

Reference epoch: 2026-01-25T2200

Science goal

- Detect variable / transient compact radio sources (including ESEs)
- Daily-cadenced forced photometry on hourly-epoch mosaics
- Light curves + Mooley variability metrics (η / V_s / m)
- Not a single deep all-sky map – a time-domain survey on a Dec strip

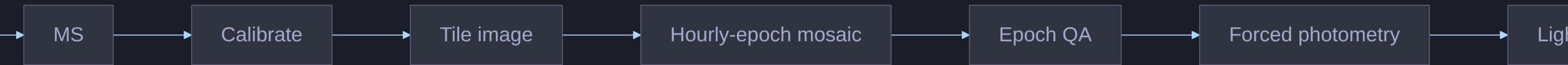
What “done” means

- Array dwells on a fixed Dec strip; sky drifts through the beam
- ~5-minute tiles → coadd into ~1-hour hourly-epoch mosaics
- Adjacent epochs overlap so flux stays continuous across boundaries
- Per-source forced photometry → variability / ESE search
- Not a 24-hour “daily mosaic”

Instrument snapshot

- DSA-110 at OVR0 – meridian drift-scan transit array
- ~96 active antennas of 117 elements in the HDF5 metadata
- L-band 1.31–1.50 GHz · 16 subbands · 187.5 MHz total
- Integration ~12.9 s · ~24 samples per ~5-min tile
- Current production Dec strip ~16.1°

Data flow



Strict QA can stop the chain before photometry

Calibration status

- Bandpass + gain tables (.b / .g) before imaging
- Acquisition order: same-date → primary calibrator → bright-source fallback → borrow nearest validated
- **Fail loud** – no silent unknown provenance
- Borrowing requires Dec-strip compatibility via provenance sidecar
- Pre-applycal RFI flagging recently wired into day/batch paths

Tile imaging status

- WSClean · 4800×4800 px · cell ~3" (tile); mosaic products often 6" pixels
- Phaseshift → applycal → image; FIELD phase/reference dirs synced
- EveryBeam needs TELESCOPE_NAME = DSA_110 after SPW merge
- Reference check: 3C454.3 recovers ~12.5 Jy/beam on test data
- These invariants are silent-failure modes if skipped

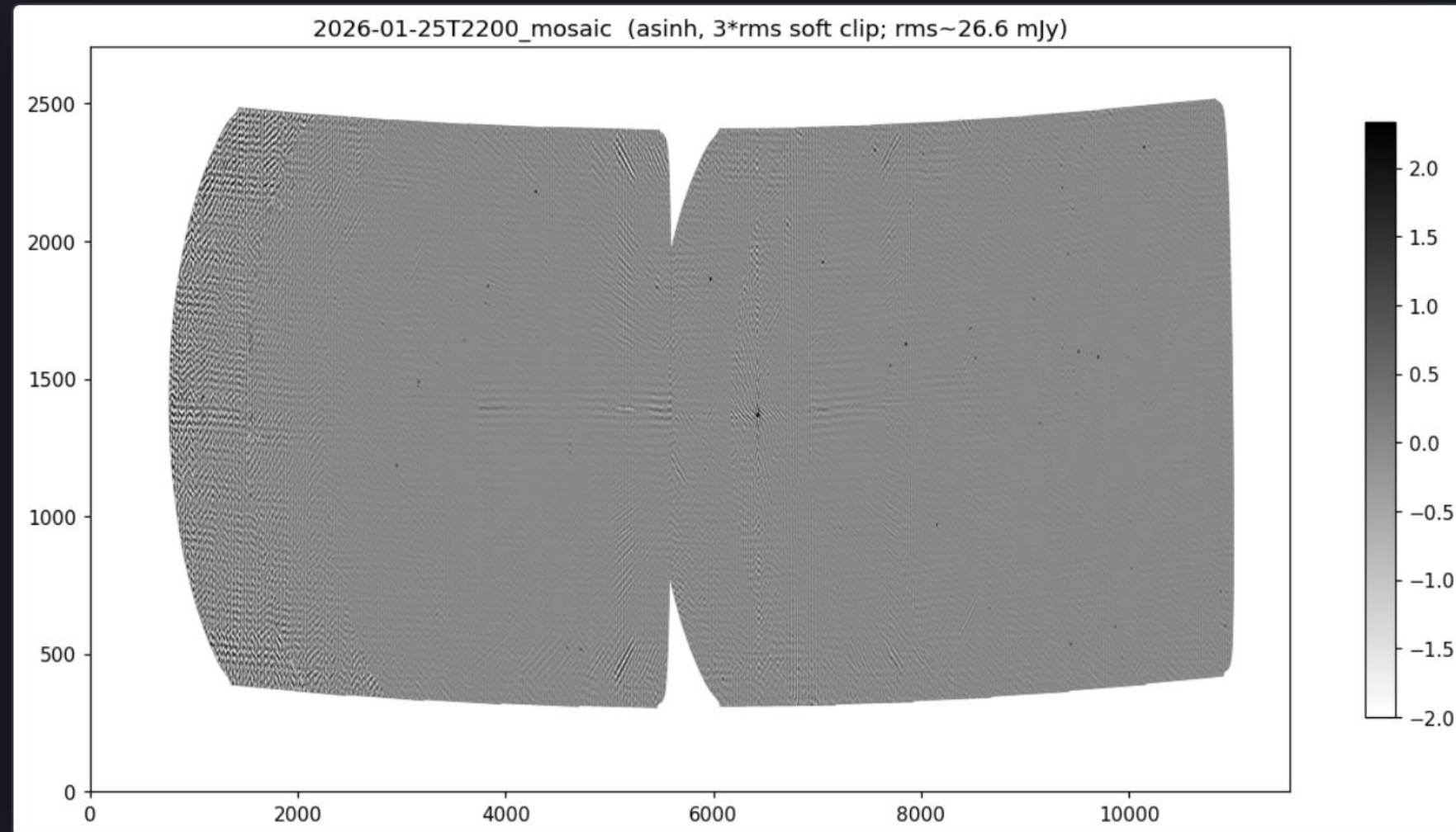
Mosaicking status

- Production: batch UTC-hour bins with ± 2 tile overlap (~20 min)
- Quicklook image-domain coadd (PB-weighted; beam-map blanking)
- ~12 tiles per hourly-epoch mosaic along the Dec strip
- Science/Deep visibility-domain joint imaging exists but is not the default product
- Sliding-window mosaics are the streaming design target

Reference product – 2026-01-25T2200

Hourly-epoch mosaic · Dec strip $\sim 16.2^\circ$ · used for all validation below

Full field



Zoom

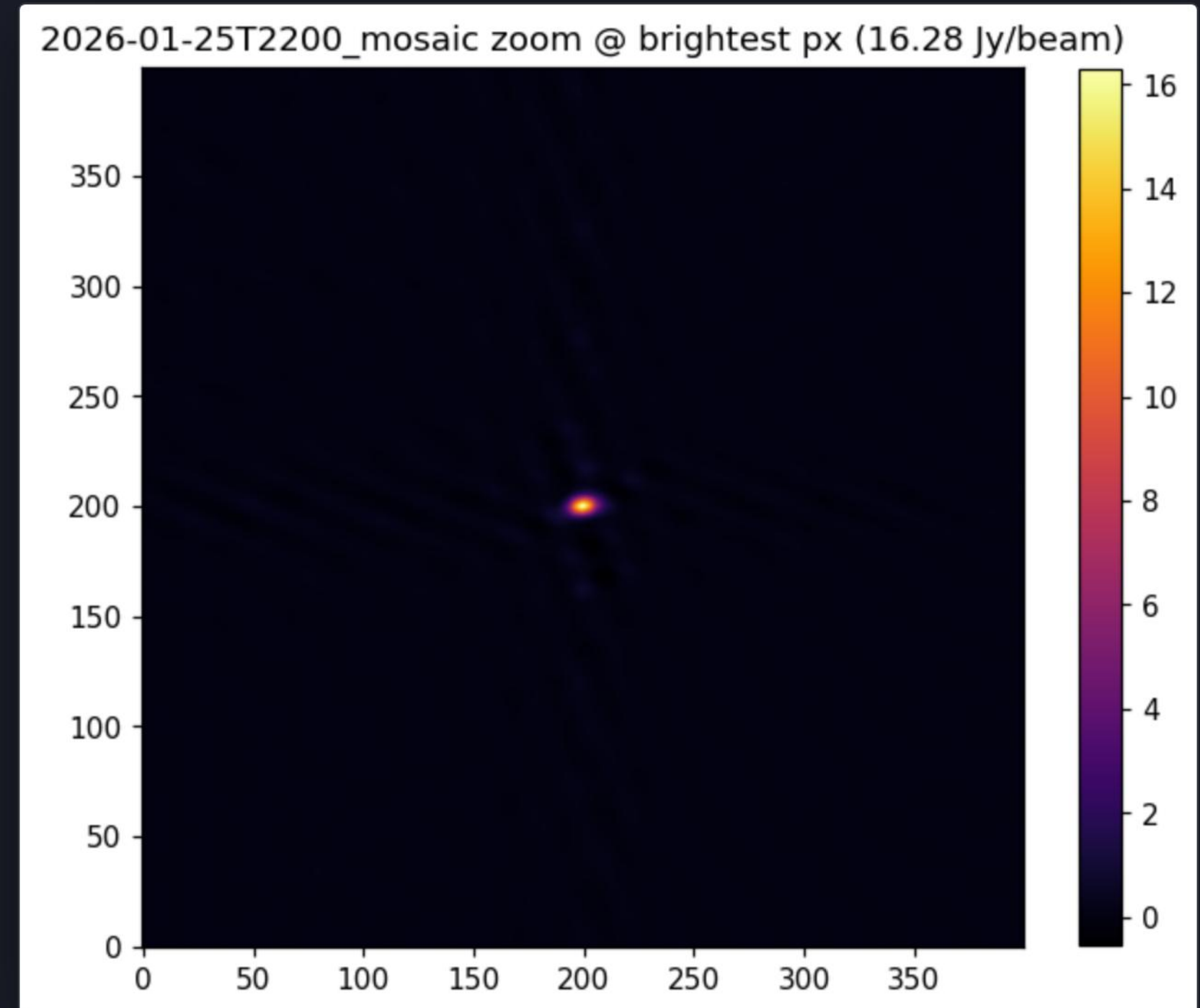
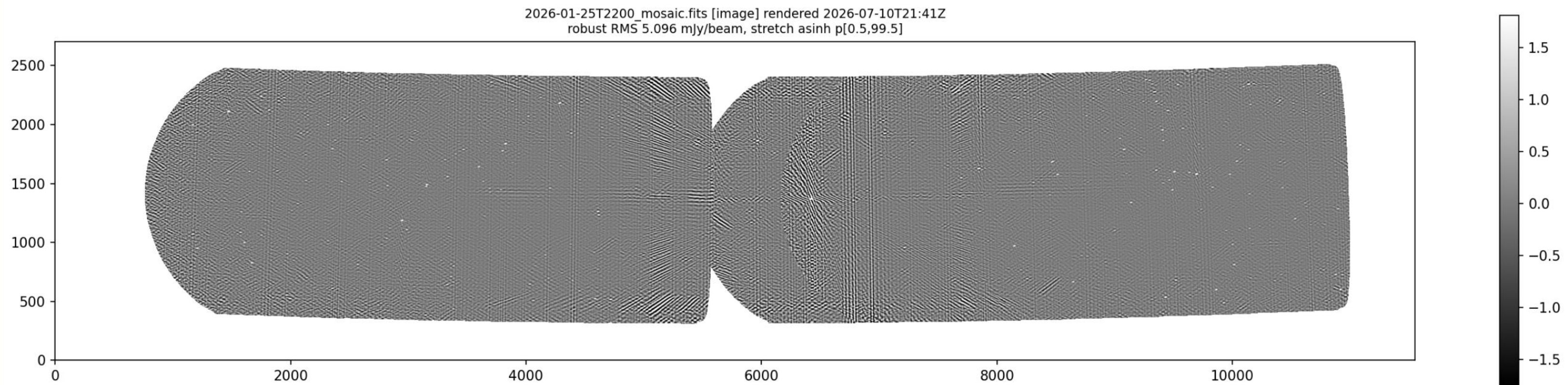


Image quality – RFI rebuild

- Pre-applycal RFI flagging cut noise substantially on this epoch
- Global robust RMS: 8.92 → 5.10 mJy/beam (~-43%)
- Bright-source peak flux essentially preserved (~16.2 Jy)
- Residual crosshatch / PSF-lattice texture remains
- Central coverage pinch (missing tile) still present
- Amplitude improved; failure topology of some image-QA metrics unchanged

RFI-cleaned mosaic

Same epoch after RFI rebuild – quieter, but lattice + coverage pinch remain



Epoch QA – three gates

- Flux scale – bright anchors vs expected
- Catalog completeness – fraction of expected sources recovered
- Noise floor – mosaic RMS / effective noise
- All three must pass for epoch **PASS**
- Strict QA (default): QA-FAIL epochs **skip forced photometry** and do not archive as science-clean

QA on the reference epoch

Gate	Result	Notes
Flux scale	PASS	Bright anchors OK
Noise floor	PASS	After RFI rebuild
Completeness	FAIL	172/383 $\approx$ 45% (need $\geq$ 60%)

- Overall epoch QA: FAIL → photometry gated off
- Coverage-corrected investigation $\approx$ 71% – not yet the production gate

Astrometry – PASS

Survey	N	Median Δ	RMS Δ	Verdict
NVSS	176	1.56"	4.69"	PASS
FIRST	16	2.22"	2.98"	PASS
RACS	215	1.86"	4.58"	PASS

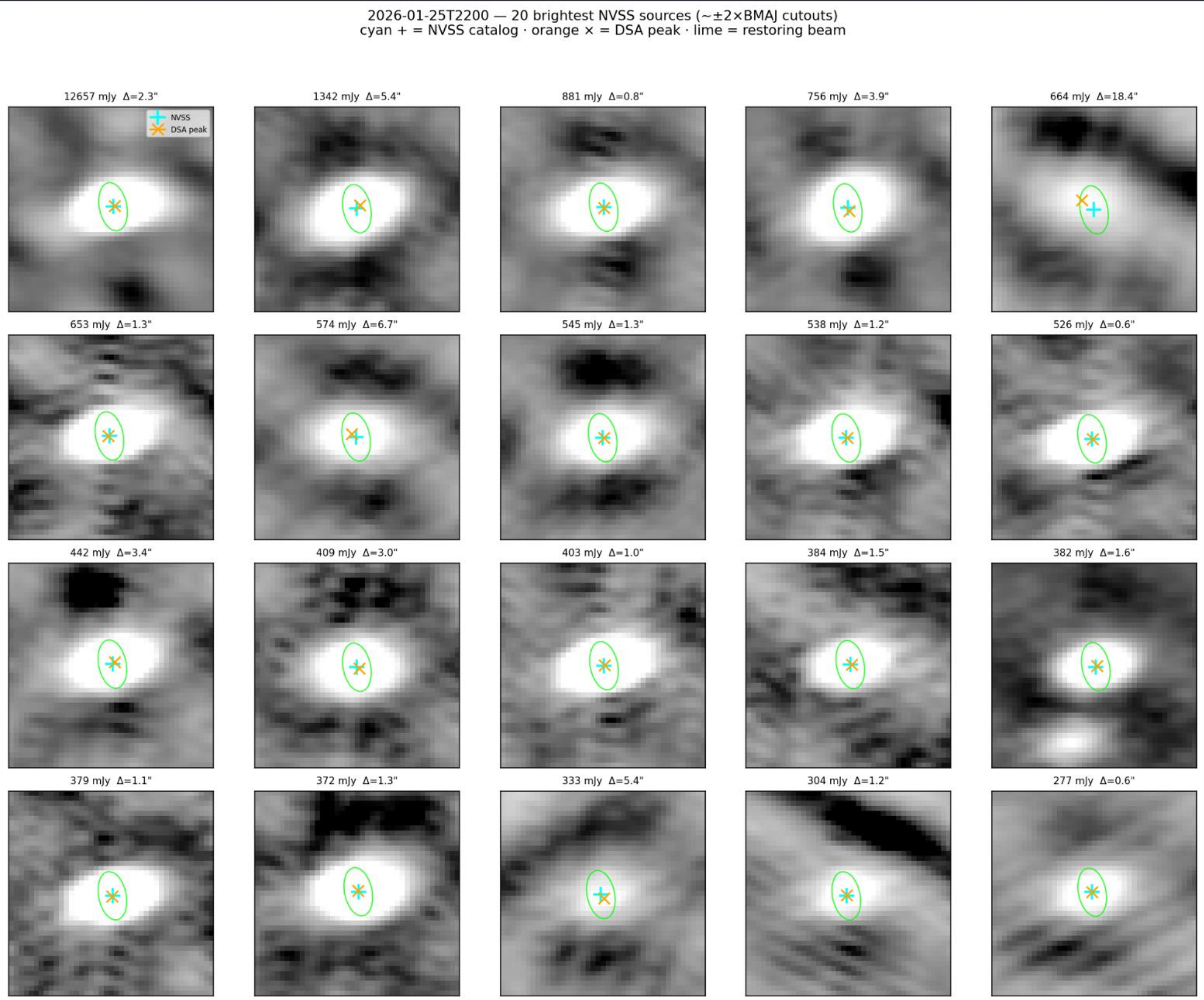
→ Gate: seeded RMS $\leq \sqrt{(BMAJ \times BMIN) / 5} \approx 8.82"$

→ Beam on this mosaic: 58.8" × 33.1"

→ Independent agreement across three surveys

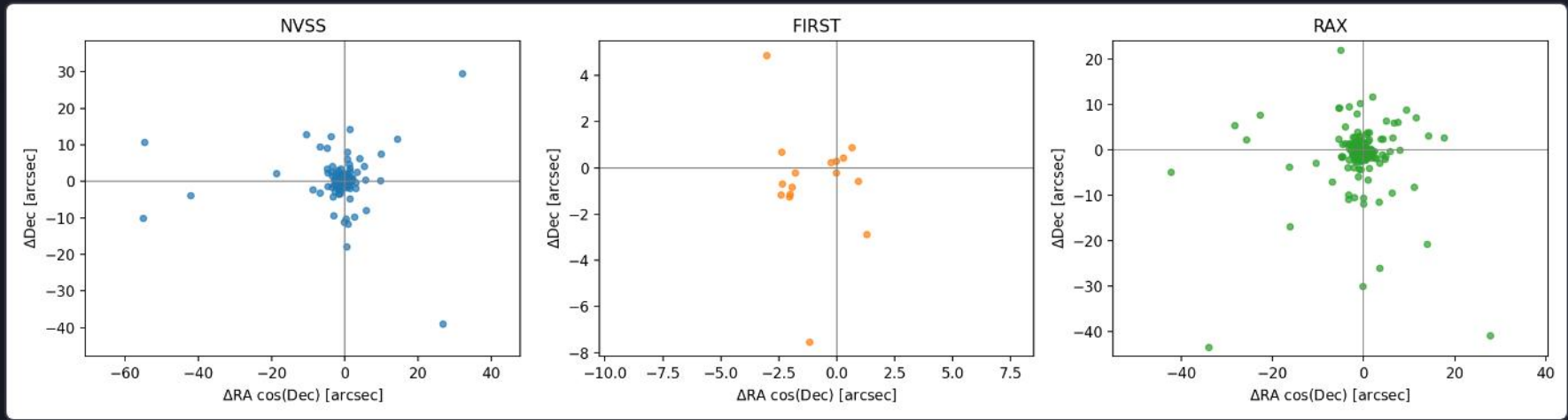
Astrometry evidence – NVSS cutouts

20 brightest NVSS · cyan + = catalog · orange x = DSA peak · lime = beam

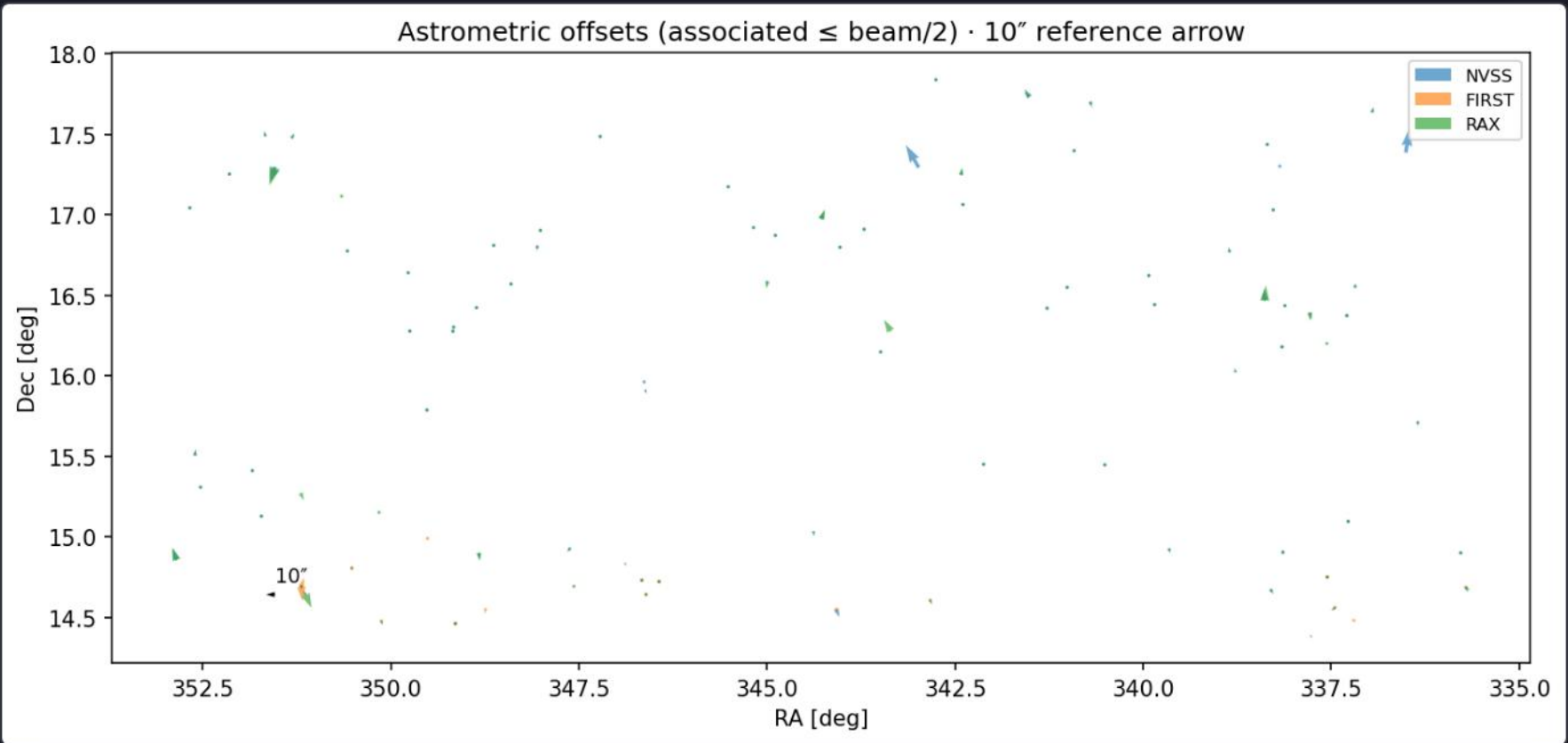


Astrometry evidence – offsets on sky

ΔRA vs ΔDec



Quiver (10" ref.)



No strong radius-dependent failure once the quiver is scaled correctly

Photometry & variability

- Condon matched-filter forced photometry is implemented
- Mooley η / Vs / m and multi-epoch pairing exist in-package
- ESE candidate scoring paths exist
- Not run on this reference epoch under strict QA
- Machinery is ahead of unblocked science products

Catalog caveat

- Forced-phot “master” catalog build claims NVSS/FIRST/RAX/VLASS inputs
- In practice, master rows are 100% VLASS today
- Astrometry validation therefore queried survey DBs directly
- Master rebuild is required before trusted multi-survey forced photometry
- This is a known Phase-2 follow-up from the astrometry design

Solid vs not yet

Solid

- HDF5 → MS → cal → tile path
- Hourly-epoch Quicklook mosaics
- Astrometry on reference epoch
- RFI amplitude improvement
- Strict QA gating discipline

Not yet

- Completeness gate (production)
- Forced phot on this epoch
- Trusted multi-survey master catalog
- Science/Deep as default product
- ESE results as science deliverable

Near-term plan

1. Fix / adopt coverage-aware completeness in epoch QA
2. Unblock forced photometry on QA-PASS epochs
3. Rebuild master catalog with real NVSS/FIRST/RAX flags
4. Promote astrometry check into routine mosaic QA
5. Later: science/deep coadds; streaming sliding-window mosaics

Takeaway

Imaging and astrometry look **credible** on the reference epoch.

Variability science products are **not ready yet** until completeness / photometry clear.

Spec: docs/superpowers/specs/2026-07-11-pipeline-status-slide-deck-design.md

Artifacts: outputs/pipeline-status-2026-07-11/